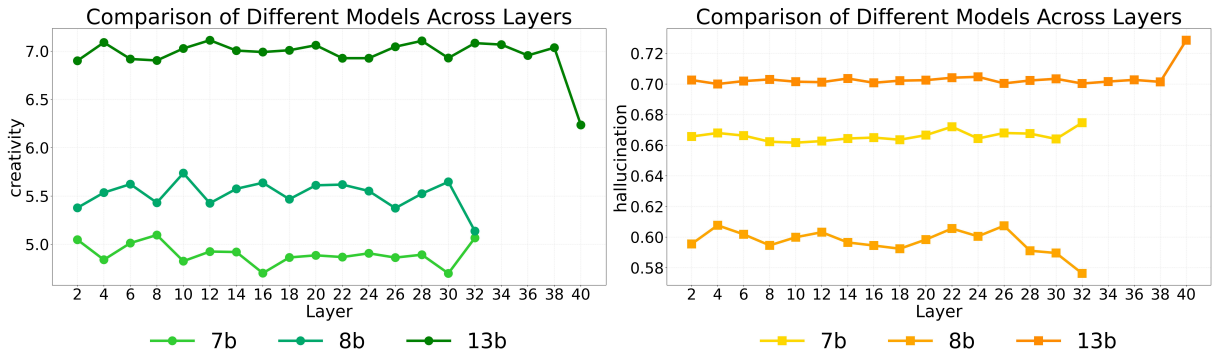

Figure 5: The left figure illustrates the creativity scores across different models, while the right figure presents the hallucination levels for the same models. Both evaluations were conducted with a temperature setting of 1.0. As observed, the LLaMA 2-13B model exhibits the highest creativity among all models. However, this increase in creativity also corresponds to a higher level of hallucination.

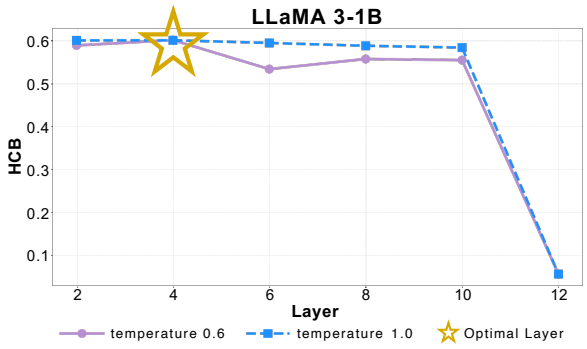

Figure 6: This figure presents the HCB score of the LLaMA3.2-1B. It is evident from the figure that ***layer-4*** consistently achieves the highest HCB score, regardless of the temperature setting.

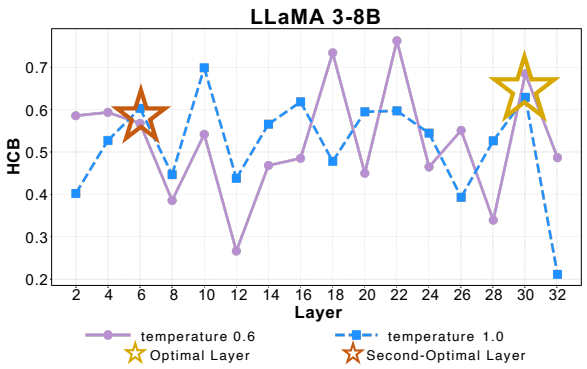

Figure 7: This figure shows the HCB score for LLaMA 3-8B. Although the results indicate ***layer-30*** is the optimal layer, we further choose ***layer-8*** to early exit considering the deeper layer causes lower efficiency.

a higher degree of unpredictability, leading to a higher risk of fabricating details that deviate from factual correctness.

These findings underscore the dual-edged nature of language models. While larger models unlock greater generative potential, they require more robust control mechanisms to mitigate hallucinations.

4.3 Investigate an Optimal Decoding Layer for Early Exit

In this part, we aim to answer whether there is an optimal decoding layer that achieves the best trade-off between creativity and hallucination, as quantified by our HCB metric. Although conventional approaches typically rely on the final layer's output, our findings suggest that earlier layers are more likely to produce responses that better balance hallucination and creativity. By skipping the later layers and selecting outputs from these relatively optimal layers, models can not only be more efficient, but also achieve an optimal balance between hallucination and creativity during generation.

The output from the final layer is not necessarily the best from a creativity perspective. Another key finding from our HCB framework is that final layers, i.e., *layer-12* of LLaMA 3.2-1B, *layer-32* of LLaMA 2-7B, and *layer-40* of LLaMA 2-13B, do not always generate the most creative responses. While the final layers refine the model's predictions and improve factual consistency, they often restrict generative flexibility, leading to more deterministic and conservative outputs. In contrast, responses extracted from mid-depth layers tend to exhibit greater creative variation while still maintaining a certain level of factual coherence. As the results shown in Figure 6, 7, 8, 10, final layer optimization is not necessarily the best strategy and does not always yield superior performance, particularly in applications that prioritize novelty and diversity over absolute factual correctness. Traditional decoding strategies often assume that final

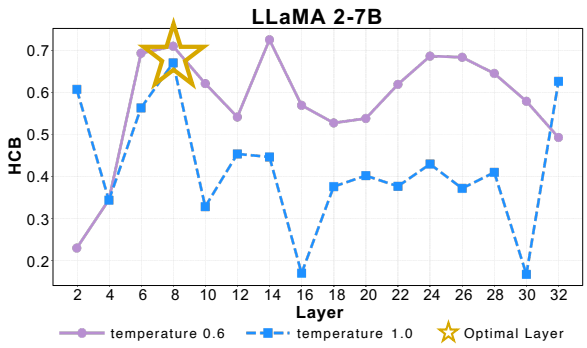

Figure 8: This figure illustrates the HCB score of the LLaMA-7B model across its layers. From the results, we can observe that ***layer-8*** emerges as the optimal layer, whether it is temperature 0.6 or 1.0.

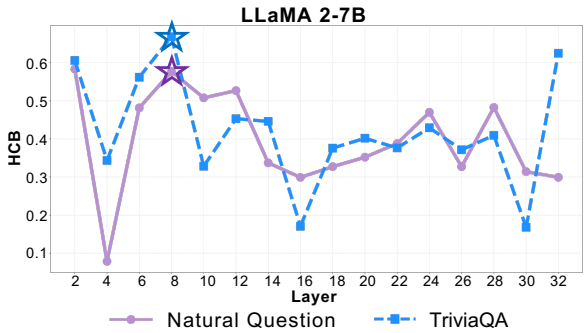

Figure 9: Illustration of the HCB score conducted on LLaMA-7B model at t = 1.0 on TriviaQA and NQ datasets. The results indicate that ***layer-8*** consistently emerges as the optimal layer for balancing creativity and hallucination in LLMs across both datasets.

layers generate superior responses, but this assumption may need to be revisited and adjusted to better accommodate creative tasks such as storytelling, poetry, and open-ended dialogue generation.

The optimal layer remains consistently effective across different temperatures and datasets, though it is not always the absolute best choice. Interestingly, our analysis reveals that each model typically has an optimal layer that maintains a stable performance under both temperature 0.6 and 1.0. For instance, in LLaMA 2-7B, *layer-8* consistently balances creativity and factual accuracy across different tasks and temperature settings, despite not being the highest-scoring layer at temperature 0.6. In LLaMA 2-13B, *layer-4* exhibits a stable trade-off between creativity and factual precision. Although *layer-30* is identified as the optimal layer in LLaMA 3-8B, its relatively deep position raises efficiency concerns. Considering the preference for earlier-layer outputs, we suggest *layer-6* for early exit.

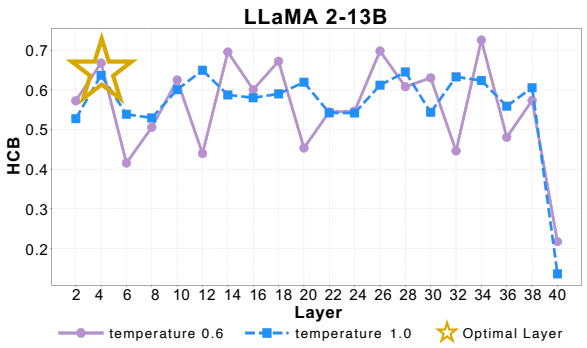

Figure 10: This figure displays the HCB score of the LLaMA-13B model. The results suggest that ***layer-4*** is the optimal layer since it remains nearly optimal when the temperature changes.

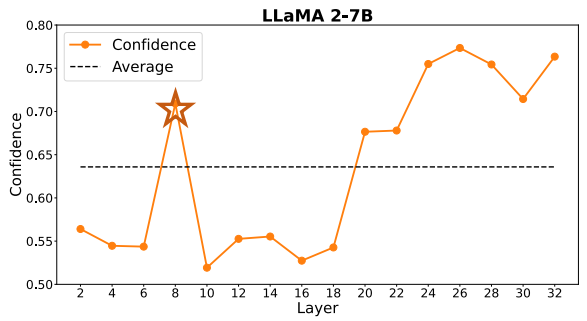

Figure 11: This figure illustrates the variations of confidence across different layers of LLaMA-7B on the TriviaQA dataset. Although the early layers show generally low confidence, there is a sharp peak at ***layer-8***, demonstrating our selection on the optimal layer.

It is worth noting that beyond temperature variations, we further analyzed the performance of LLaMA 2-7B on the TriviaQA and NQ datasets, as illustrated in Figure 9. The results demonstrate that the optimal layer in terms of the HCB metric remains consistent across different QA datasets, i.e., *layer-8* remains the one that optimally balances the tradeoff between hallucination and creativity in LLMs. The pattern shown in Figure 11 further supports the idea that *layer-8* is a key decision-making layer in the model. This further demonstrates that the identified optimal layer is not only specific to a given model but also has broader generalizability across common QA datasets, verifying the robustness of our HCB-based selection.

5 Conclusion

This paper reviews the development of hallucination and creativity in LLMs and proposes a hierarchical evaluation framework, HCL, to explore their interaction across different layers. Additionally,

we identify the optimal layer that best balances the tradeoff between hallucination and creativity in LLMs. We have conducted extensive experiments to find key factors influencing both aspects. This study provides a quantitative definition of creativity and offers valuable insights for further exploration of LLM performance across different tasks.

Ethics Statement

Our proposed method aims to improve the reliability and creative capabilities of LLMs by analyzing and utilizing responses from different decoding layers. While HCL has the potential to reduce hallucinations while preserving creativity, it is essential to acknowledge the ethical implications associated with our work from the following aspects:

- **Misinformation & Reliability:** LLMs can generate highly plausible yet incorrect information. By investigating hallucination mechanisms, our study provides insights into distinguishing between factual and misleading outputs. However, our method does not entirely eliminate hallucinations, and caution should be exercised when applying it in high-stakes scenarios such as healthcare or finance.

- **Bias & Fairness:** LLMs may inherit biases related to gender, ethnicity, and other social factors. Since our framework evaluates hallucination and creativity within existing models, it does not explicitly mitigate bias. Future research should consider fairness-aware approaches to ensure responsible AI deployment.

- **Computational Impact & Efficiency:** Our layer-wise analysis and early exit strategies aim to optimize computational efficiency, potentially reducing energy consumption in large-scale model inference. However, running extensive experiments with multiple models still requires substantial computational resources.

Limitations

The correct answer types provided by existing datasets are limited to evaluate the creativity of the LLM's generations. In addition, our framework is limited to the closed-ended question-answering domain, where a question has multiple objective ground-truth answers so that we can justify the correctness of LLM generated answer. Extensive analysis of HCL on open-ended question-answering tasks in real world scenarios is beyond the scope of the current study and is left as future work.

The current definition of creativity is relatively narrow, as it distinguishes diversity based on correctness but does not fully consider novelty and originality in subjective or open-ended tasks. In future work, we will expand the evaluation dimensions of creativity to encompass a broader range of creative expressions.

Additionally, our experiments are limited to a subset of models and do not comprehensively cover LLMs of different scales. In the future, we plan to incorporate **LLaMA 70B**(Touvron et al., 2023), **DeepSeek-R1**(Guo et al., 2025), and **GPT-4o**(Hurst et al., 2024), among other large-scale models, to further validate the applicability of the HCL framework across different model architectures and sizes.

References

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Tuhin Chakrabarty, Philippe Laban, Divyansh Agarwal, Smaranda Muresan, and Chien-Sheng Wu. 2024. Art or artifice? large language models and the false promise of creativity. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pages 1–34.

Daixuan Cheng, Shaohan Huang, Junyu Bi, Yuefeng Zhan, Jianfeng Liu, Yujing Wang, Hao Sun, Furu Wei, Denvy Deng, and Qi Zhang. 2023. Uprise: Universal prompt retrieval for improving zero-shot evaluation. *arXiv preprint arXiv:2303.08518*.

Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. 2023. Dola: Decoding by contrasting layers improves factuality in large language models. *arXiv preprint arXiv:2309.03883*.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*.

Mostafa Elhoushi, Akshat Shrivastava, Diana Liskovich, Basil Hosmer, Bram Wasti, Liangzhen Lai, Anas Mahmoud, Bilge Acun, Saurabh Agarwal, Ahmed Roman, et al. 2024. Layer skip: Enabling early